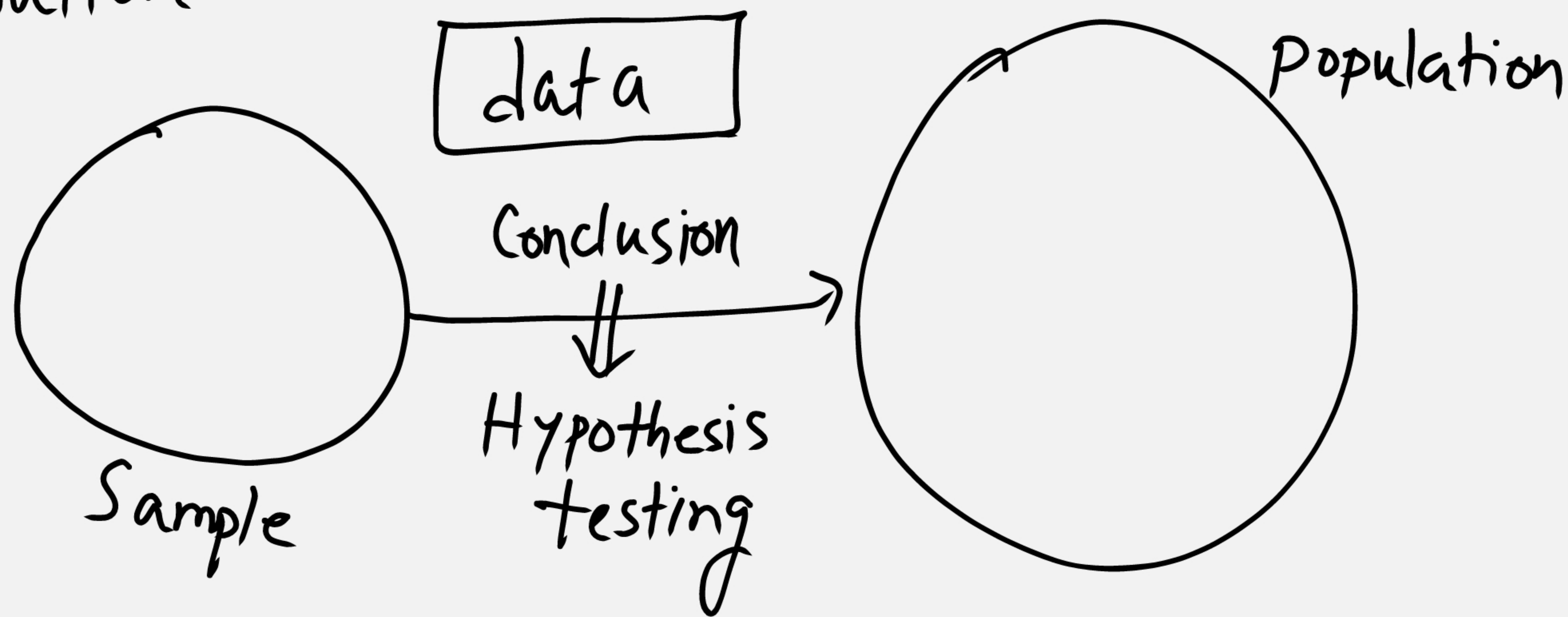


Hypothesis Testing

Hypothesis testing is a form of Statistical Inference that Uses data from a Sample to draw Conclusion about a population parameter or a population probability distribution.



Topic

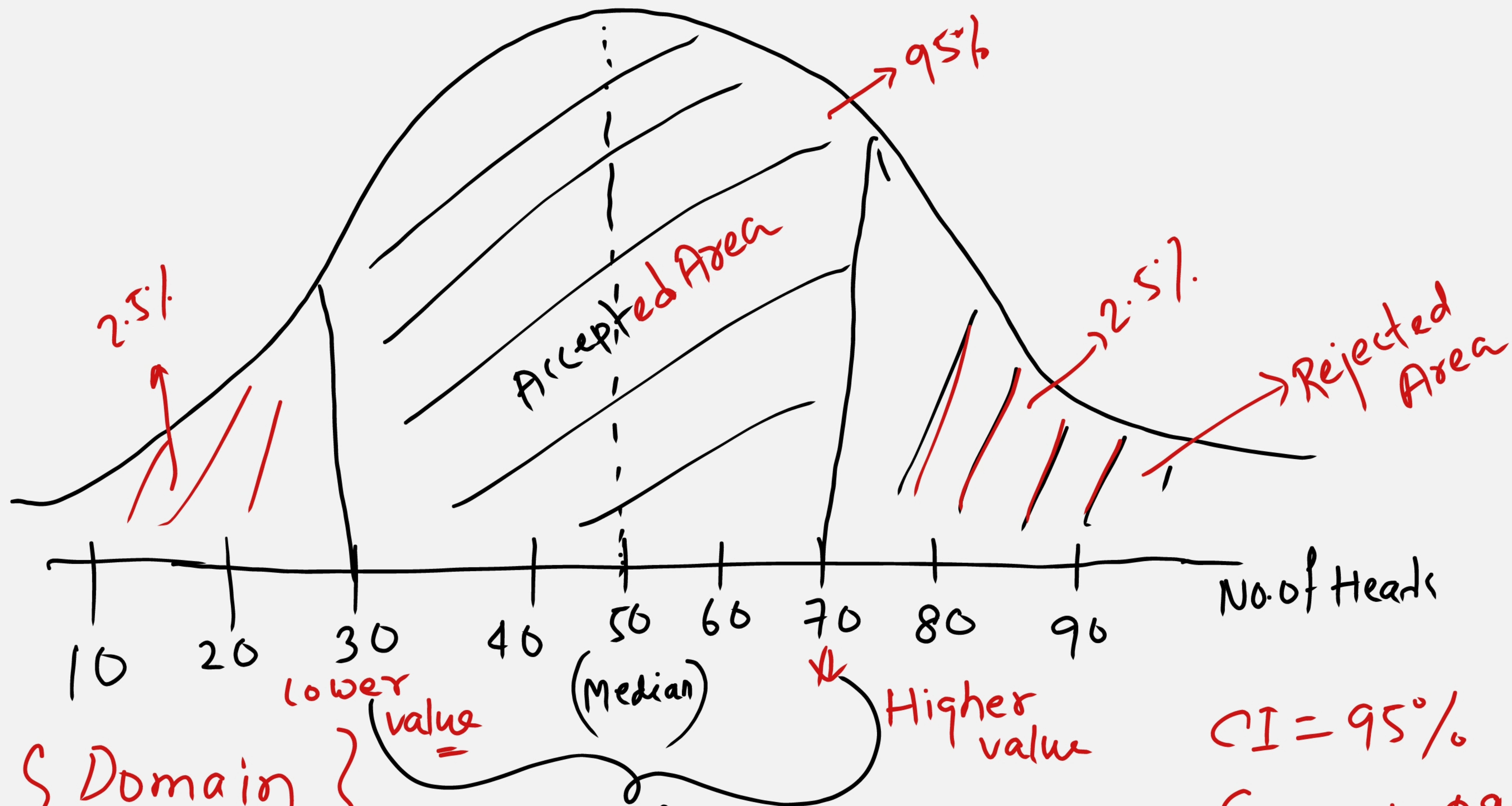
- 1) Eigenvalue & Vector
- 2) C-H Theorem
- 3) Vectors, properties and operations.

Step of Hypothesis Testing:

① Null Hypothesis (H_0): Coin is fair $\rightarrow$ Assumption

② Alternative Hypothesis (H_1): coin is not fair

③ Experiment: { Tossing a coin for 100 times } $\Rightarrow$ population



{ Domain Expert Decide }

Confidence Interval Boundary Region

$$CI = 95\%$$

$$Sv = 1 - 0.95 = 0.05\% \text{ (significance value)}$$

- ① Setup a Hypothesis Testing - H_0, H_1
- ② Significance level $\rightarrow 0.1\%, 1\%, 5\%, 10\%$ (α)
- ③ Setting a Test Criteria - Z-test $\checkmark$ t-test, χ^2 -test, $\hat{F}$ -test
- ④ Test Computation — Test
- ⑤ Making decision

The mean height of a random sample of 100 students is 64" and standard deviation is 3". Test the statement that mean height of population is 67 at 5% significant level.

$$n=100, \bar{x}=64, s=3$$

$$\mu=67, \alpha=5\%=0.05$$

$$SE_{\bar{x}} = \sqrt{\frac{\sigma^2}{n}}, \text{ SD of population}$$

$$= \sqrt{\frac{s^2}{n}}, \text{ SD of sample variance}$$

$$\checkmark H_0: \mu = 67$$

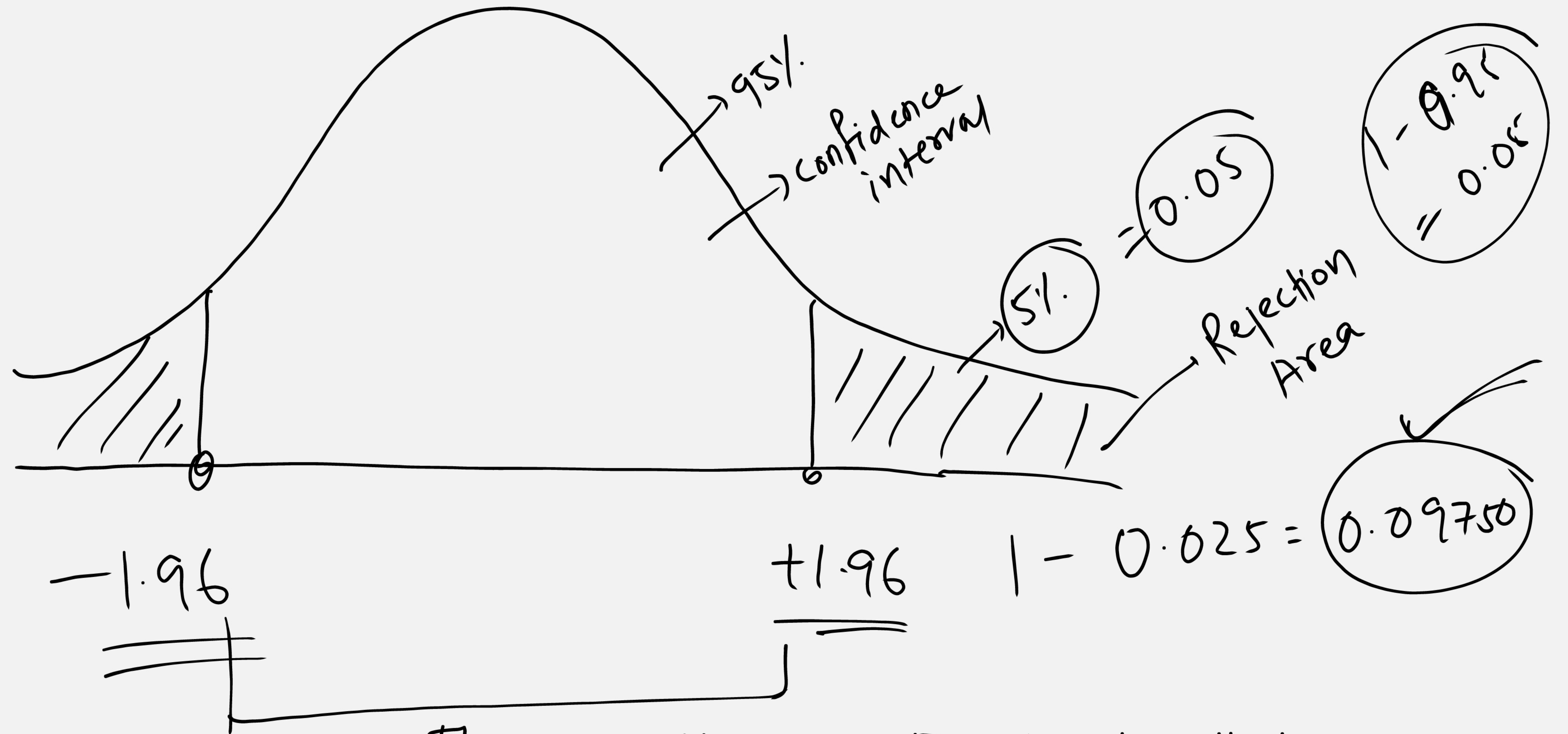
$$\checkmark H_1: \mu \neq 67$$

$$\alpha = 0.05$$

Standard Normal variate

$$|Z| = \frac{\text{difference between Population mean} - \text{Sample mean}}{\text{Significant Error } (SE_{\bar{x}})} = \frac{|\bar{x} - \mu|}{SE_{\bar{x}}}$$

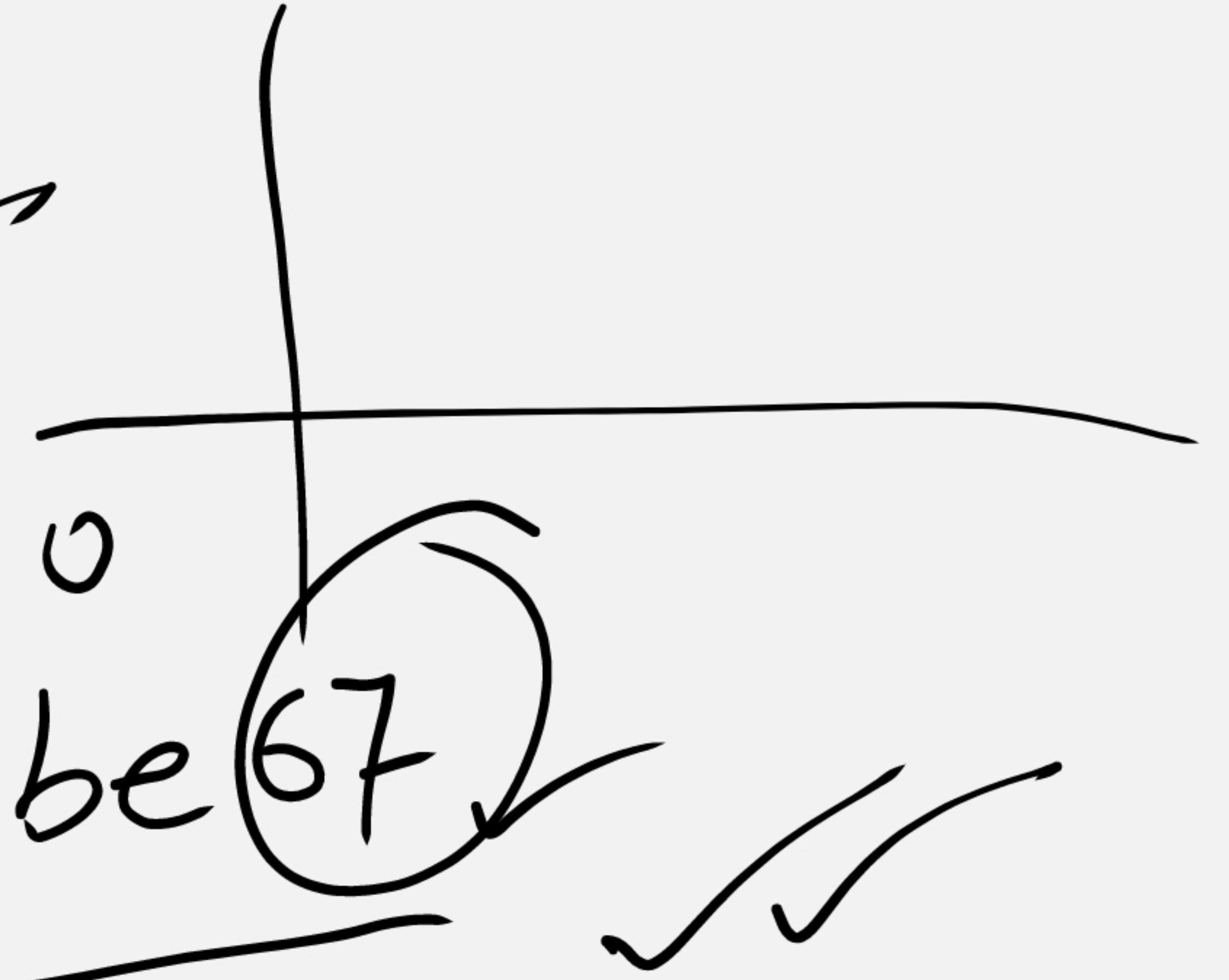
$$|Z| = \frac{|64 - 67|}{0.3} = \frac{3}{0.3} = 10, \boxed{|Z| = 10} \checkmark$$



Then we will accept the Null hypothesis,

At 5% Significance level the critical value of $|Z|$ is 1.96 ✓
Since the calculated value of $|Z| = 10 > 1.96$ at 5% Significant level.
Therefore null hypothesis is rejected. ✓

Conclusion: Mean height of population cannot be 67 ✓✓



P
 0.1
 0.05
 0.01
 0.001

Significance level
 10%
 5%
 1%
 0.1%

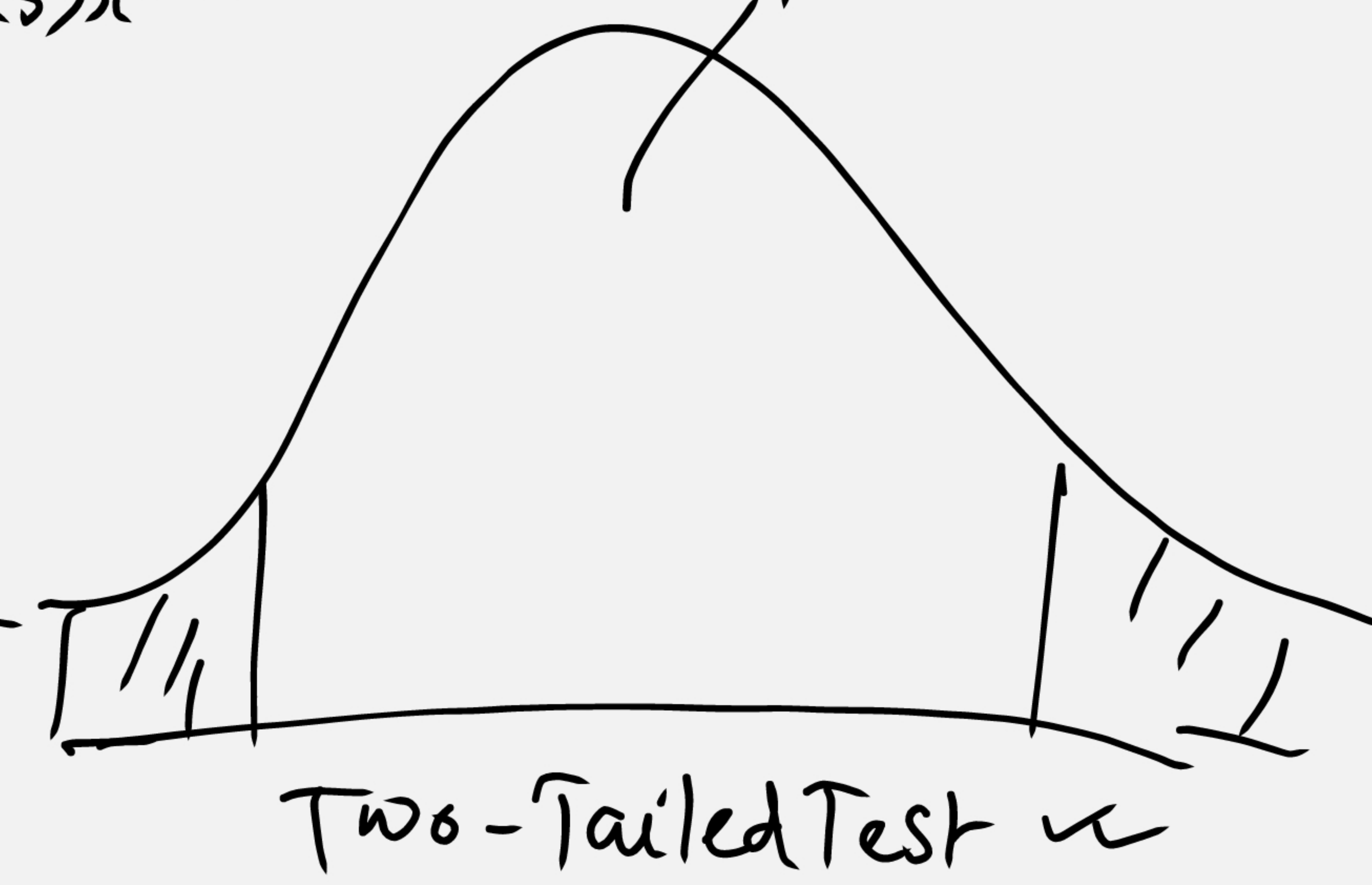
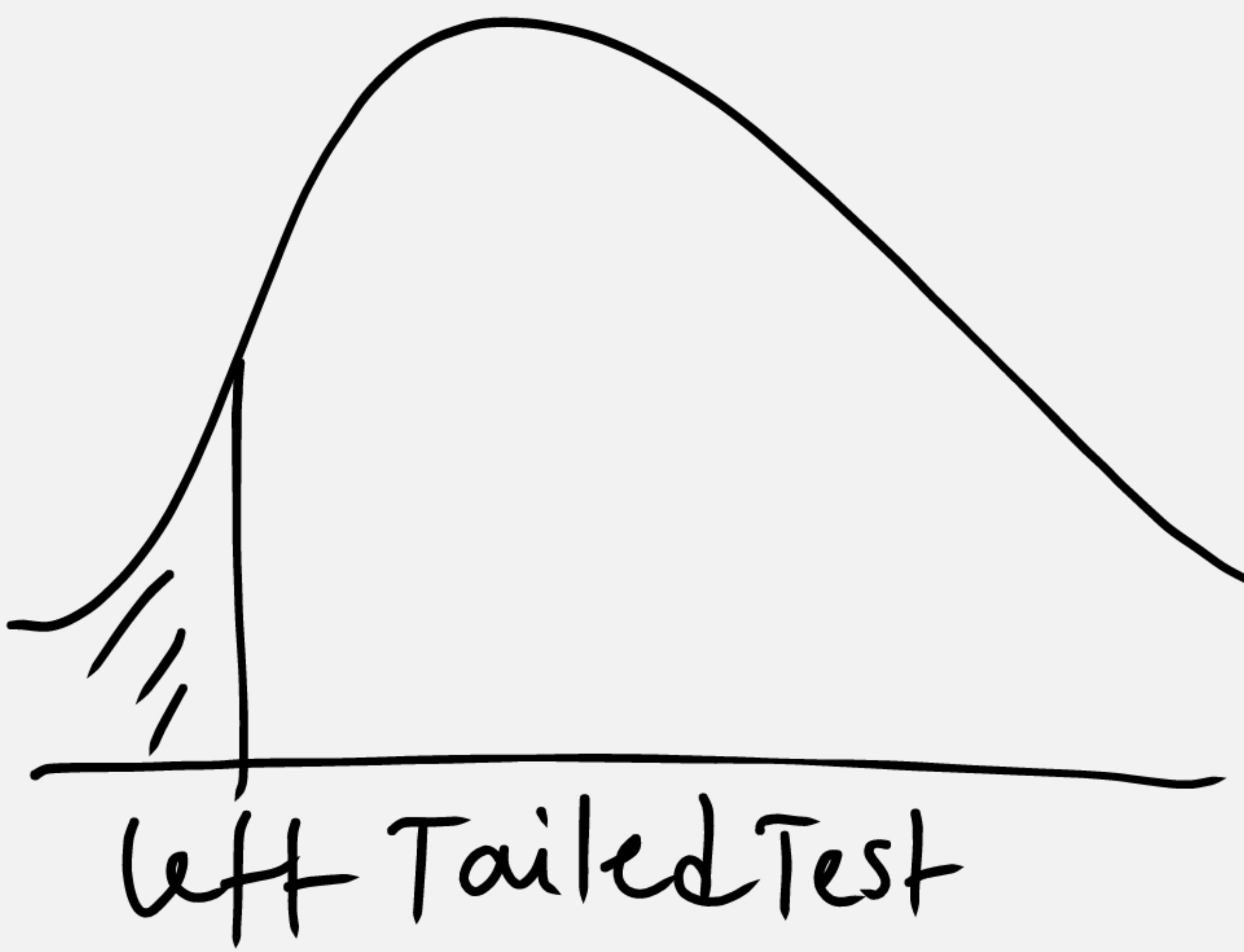
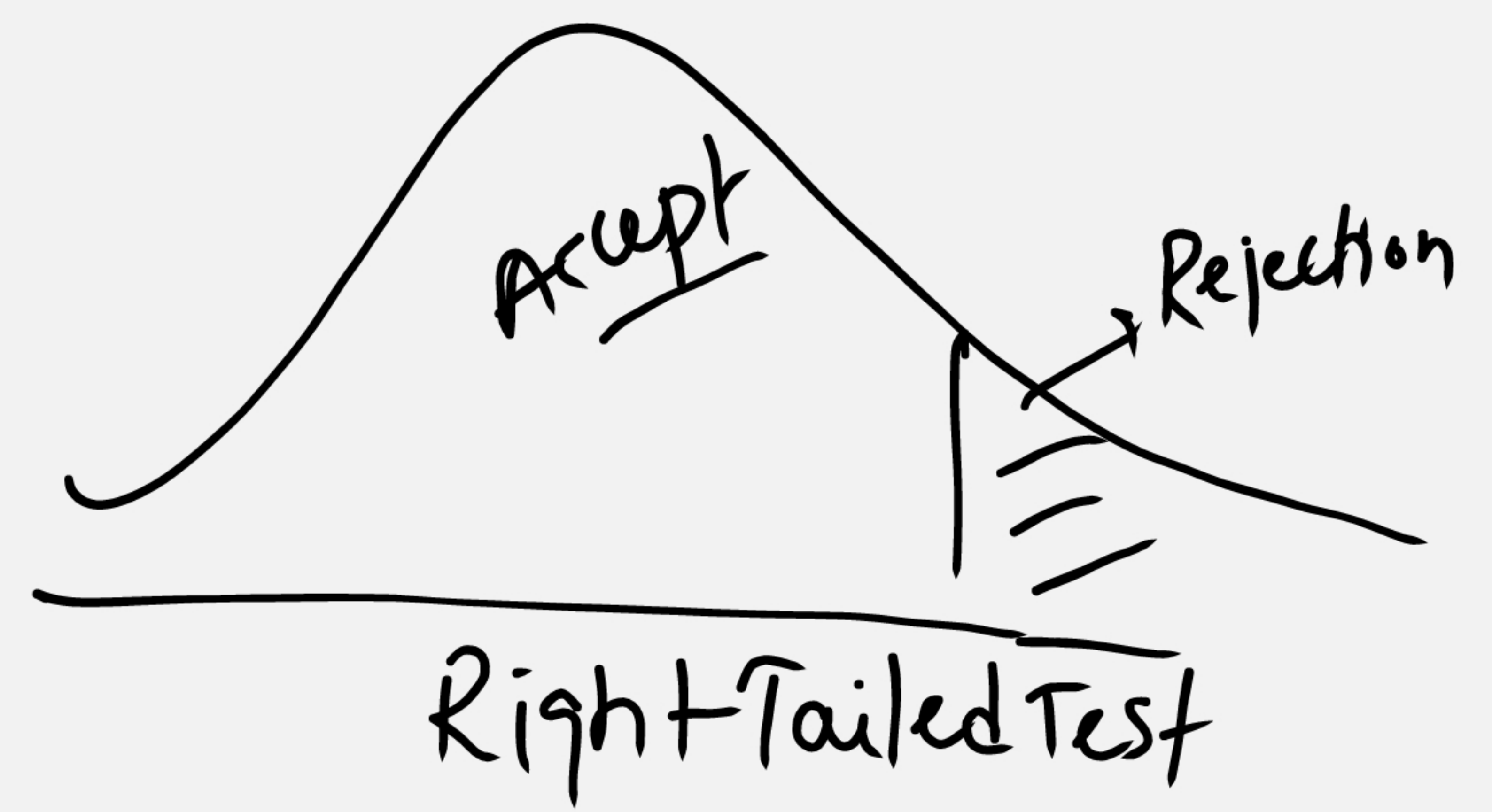
Two tailed test
 1.65
1.96
 2.58
 3.29

One-tailed Test
 1.28
1.64
 2.33
 3.10



One-Tailed Test and Two-tailed Test

H_0 : class students ≤ 90
 H_1 : class students > 90



H/W

The average heights of all residents in a city is 168 cm with a population Std $\sigma = 3.9$. A doctor believe the mean to be different. He measure the height of 36 individuals and found the average to be 169.5 cm.

(a) State null and alternative hypothesis.

(b) At 95% CI, is there enough evidence to Reject the Null hypothesis.